Stage selection — EDA

Question: given an approved value stream, how well does the model pick the lifecycle **stages** (Jira Epics) the idea card impacts — and where does it fail?

Task (distinct from VS prediction): VS prediction picks *which broad value streams* a change touches. Stage selection is narrower: for **one already-approved value stream**, choose which of its **5–15 governed lifecycle stages** the work falls into. The model picks only from that value stream's own candidate stages — it cannot invent or borrow.

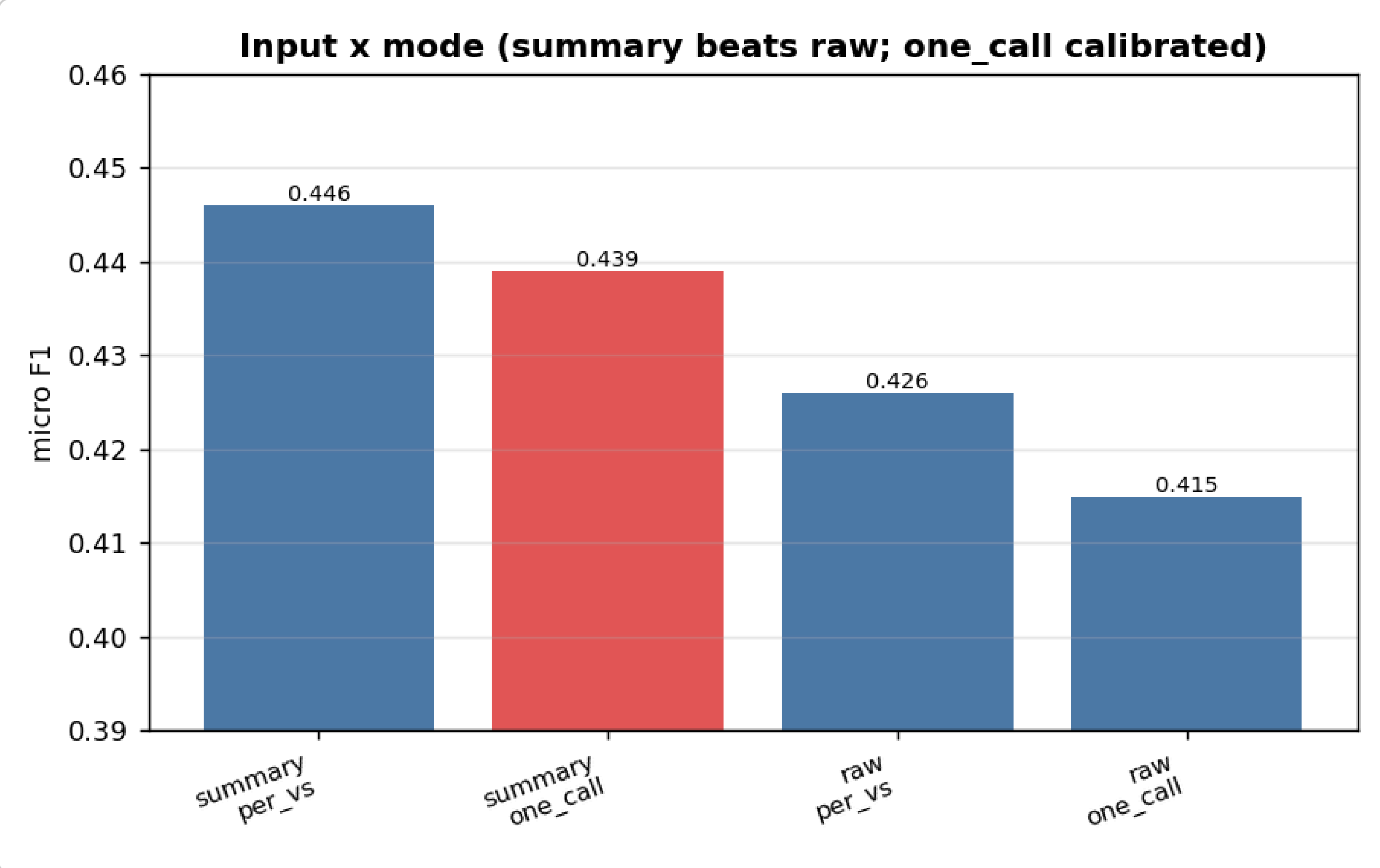
Setup

- **GT** = the stages the BA actually tagged, read straight from each Epic's **Value Stream Stage** cascading field (`<vs> {vs_id} – <stage> {stage_id}`) — authoritative, not fuzzy title matching.
- **Cohort** = tickets with ≥3 value streams and ≥5 GT stages (the discriminating multi-VS cases), plus a full-population run. `count`-free: the model returns 1–3 stages per VS, the architect trims.
- **Two modes** compared: `per_vs` (one LLM call per value stream) and `one_call` (all value streams in one batched call — the production path).
- **Two inputs** compared: the stored **summary** vs the **raw** idea card.

Terms

term	meaning
per_vs	one LLM call per value stream (N calls for N value streams)
one_call	one batched call for every value stream at once (the production path)
coverage	share of GT stage ids that are actually IN the catalogue the model picks from — the recall ceiling
mislink (cross_vs)	a stage the batched call assigned to the WRONG value stream
fallback	the resolver returned the whole lifecycle (empty/invalid LLM pick)

Finding 1 — summary beats raw (the opposite of VS)



input	mode	F1
summary	**one_call**	**0.439**
summary	per_vs	0.446
raw	per_vs	0.426
raw	one_call	0.415

For stages, the summary wins (by ~2 pts, consistent across modes, and it held after the raw input was cleanly re-labelled `ideaCard` rather than framed as a summary). This is the **opposite of VS**, where the full raw prompt won by ~7 pts. The reason: VS needs **breadth** (the implied/downstream areas live in raw detail); stage selection is a **narrow action→stage match**, and the concise summary already isolates the action — raw re-buries it in noise.

> **Raw to decide which VS · summary to decide which stages.** (Production still feeds raw for architectural consistency across theme generation; the cost is ~2 pts, well within the structural ceiling below — see Finding 5.)

Finding 2 — one_call is the right mode (calibrated, not just cheap)

`per_vs` edges F1 by ~1.7 pts, but only by **over-fetching** (avg 2.6 picks vs 1.9 GT). `one_call`:

- is **calibrated** (predicts ≈ the right count),

- has **higher precision** and **near-zero fallback** (0.005),
- and is **1 call instead of N** (~6× cheaper on a 6-VS ticket).

So `one_call` is the production path; `per_vs`'s recall edge is just overfetch you'd pay N calls for.

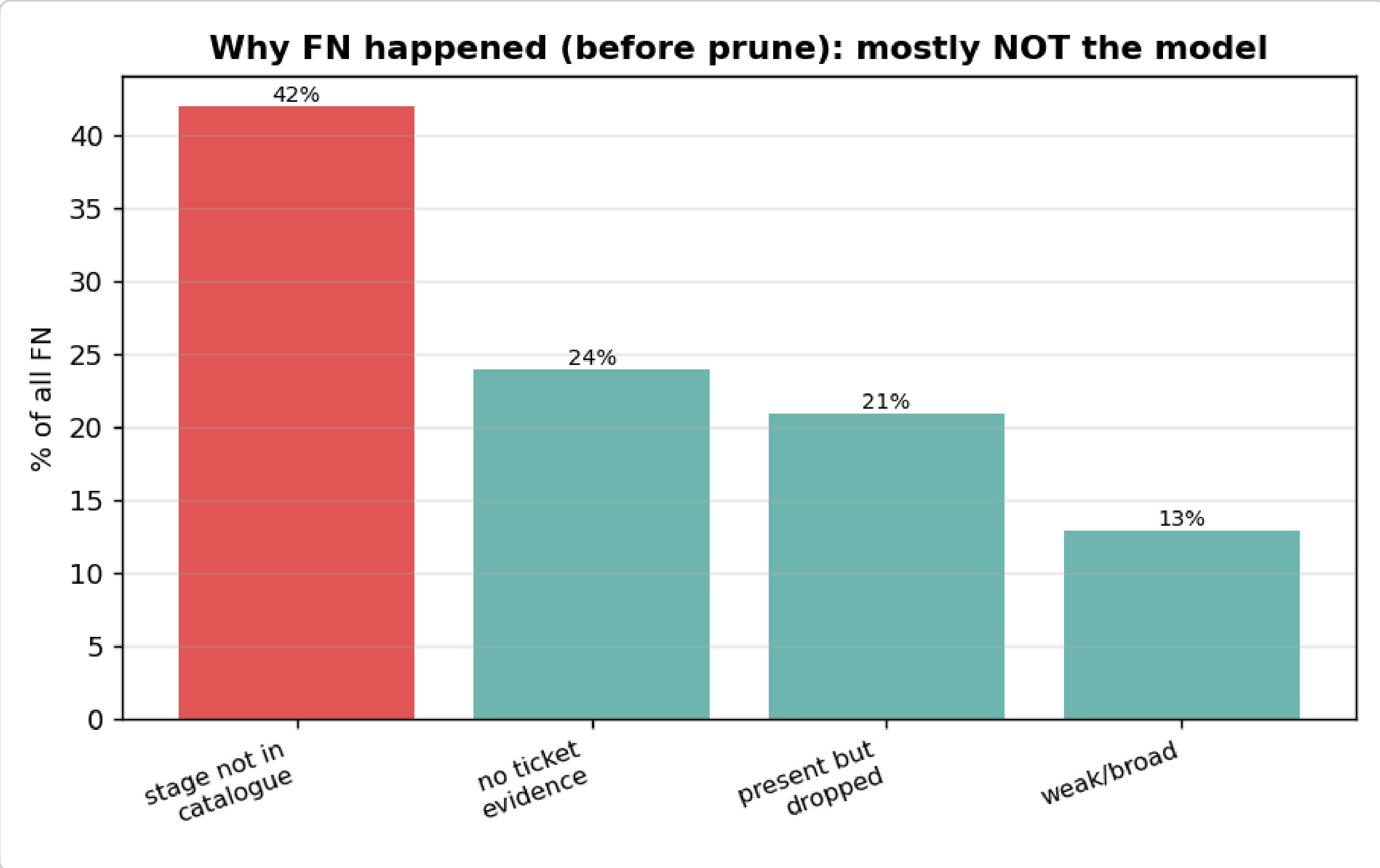
Finding 3 — interlinking solved (prevent + salvage)

The batched call can assign a stage to the wrong value stream (cross-VS mislink). Two layers fixed it:

- **Prevent** — a STRICT VALUE-STREAM ISOLATION block: candidate lists are disjoint, only return ids printed under THAT value stream.
- **Salvage** — a stage id is globally unique to one value stream, so a mislinked pick is reassigned to its true owner instead of being dropped (a safety net; ideally never used).

Result: cross_vs mislinks went from 1–5 → 0. The metric still records the model's raw behaviour, so we know prevention is working, while output gets the corrected assignment.

Finding 4 — the recall ceiling was a CATALOGUE gap, and fixing it lifted the fair numbers



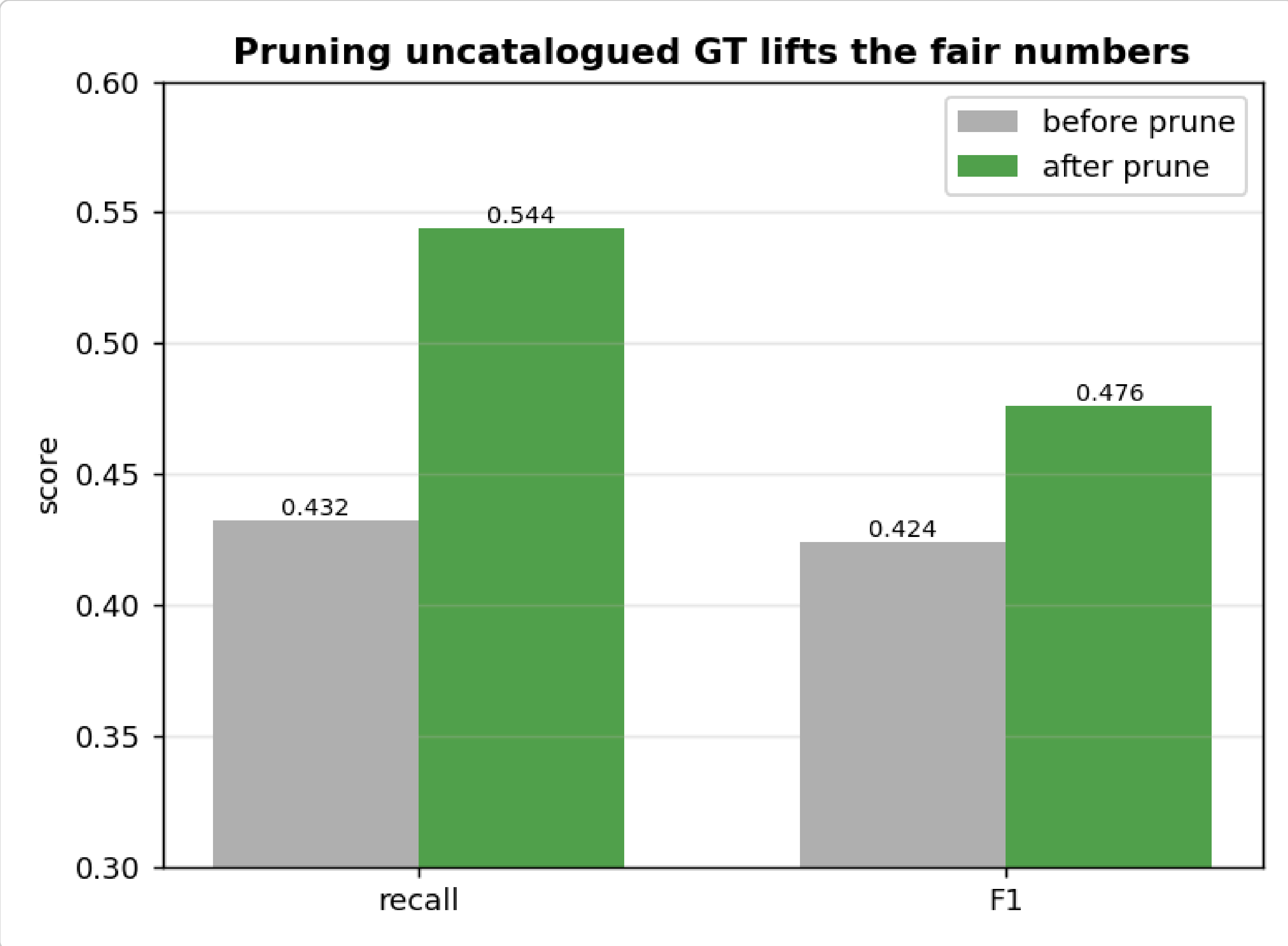
The first runs showed recall ~0.43 against a **76% coverage** ceiling. Diagnosis (no LLM) split the false negatives by cause:

- **~42% of FN: the GT stage id wasn't in the catalogue at all** — a valid `VSS#####` stage the BA tagged in Jira, but the catalogue export never listed it (e.g. *Determine Corrective Action*, *Train Provider*). The model

cannot pick what isn't an option. It was *not* a VS-mapping issue — all 44 GT value streams were present; only specific stages were missing.

- **~24% of FN: the stage is a candidate but the ticket has no evidence for it** — the BA chose it from lifecycle convention / outside knowledge (label noise).
- ~21% present-but-dropped, ~13% weak/broad.

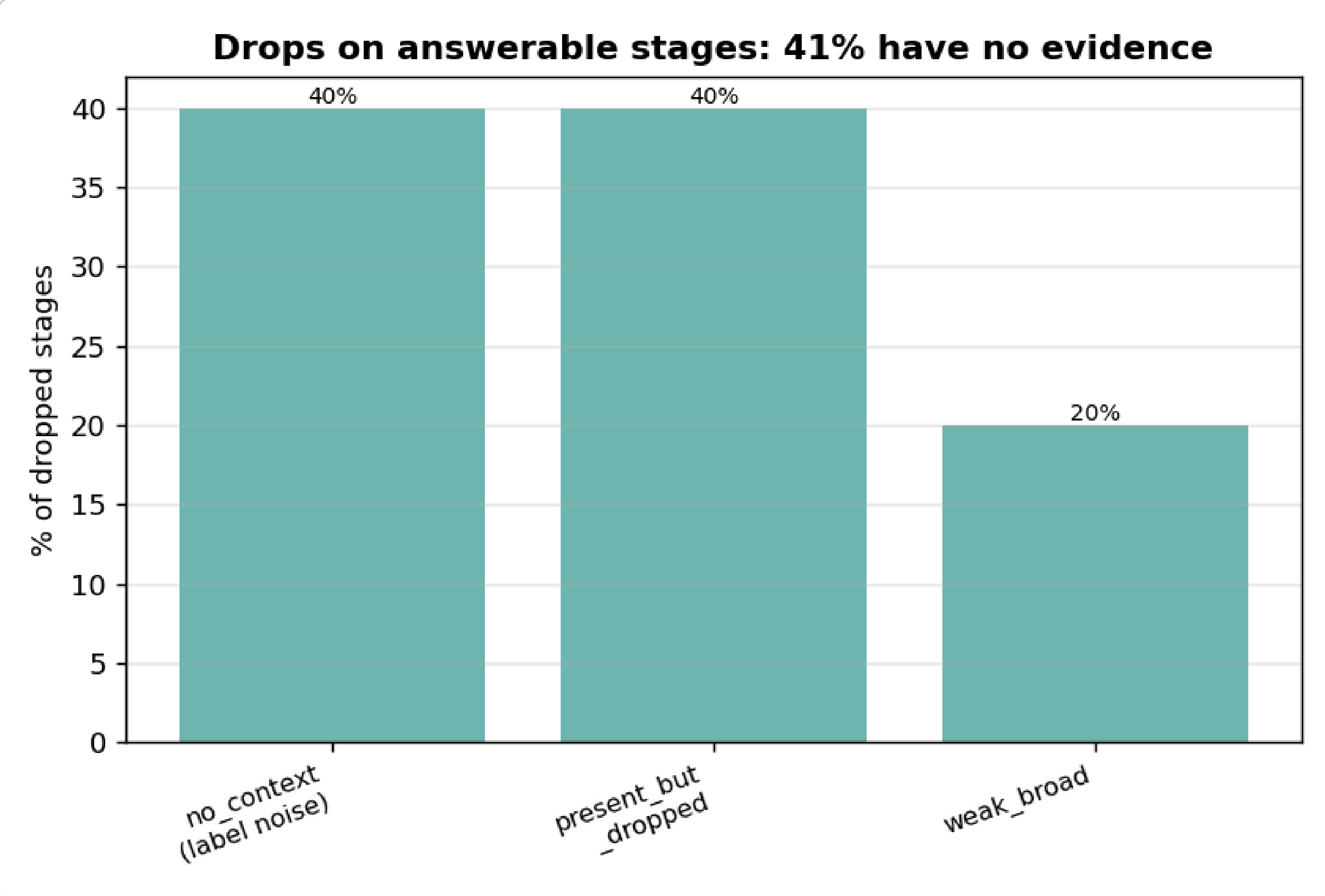
So **~66% of FN was NOT the model's fault** (un-pickable or un-derivable). The fix: **drop the uncatalogued GT stages** (retired / out-of-catalogue) at extraction — the model can't be graded on stages that aren't options.



metric (raw / one_call)	before prune	after prune
recall	0.432	**0.544**
F1	0.424	**0.476**

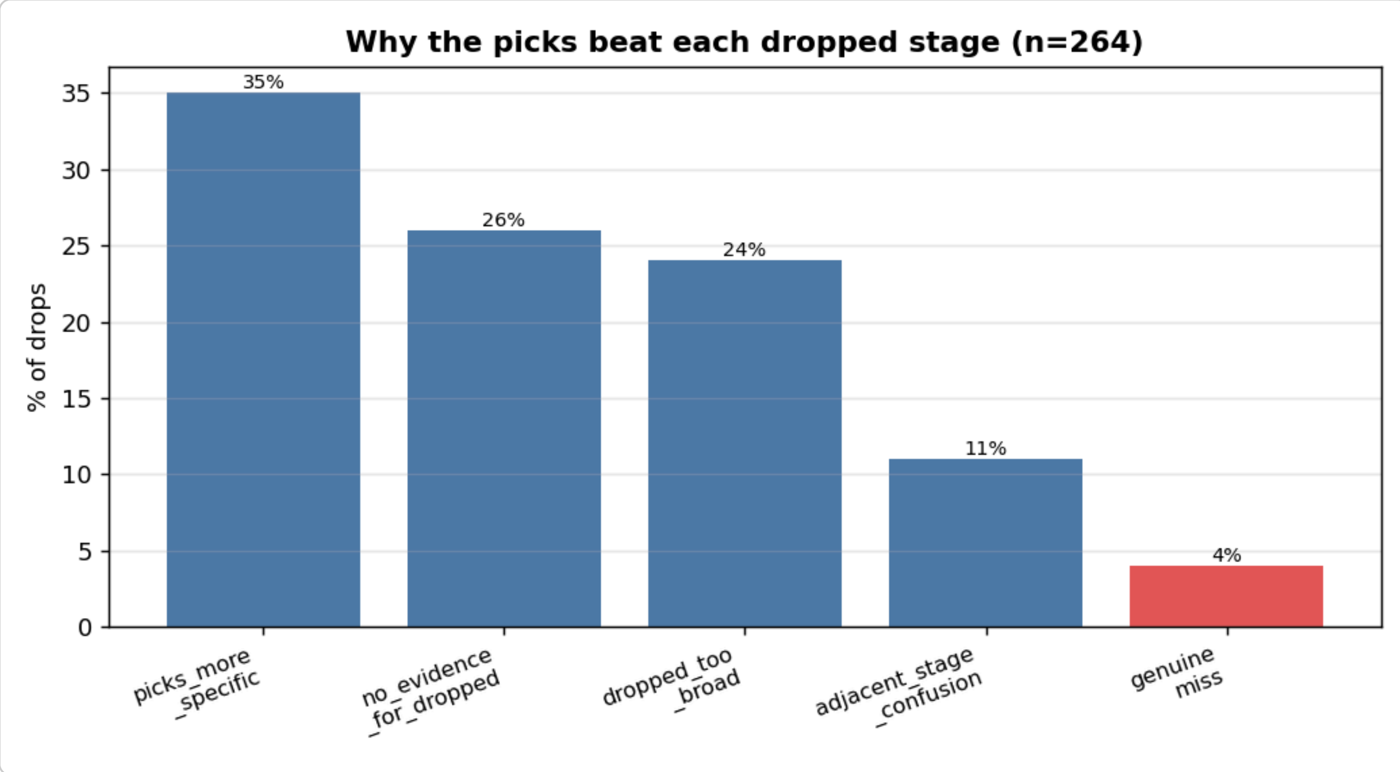
Removing the ~18% un-pickable GT lifted recall **+0.11** and F1 **+0.05** — a pure data fix, zero model change. The post-prune numbers are the **fair baseline**: the model against *answerable* GT.

Finding 5 — what's left is mostly label noise + defensible narrow picks



On the answerable stages (post-prune), the drops break down as (n = 259 classified):

grounding	count	share	meaning
no_context_for_stage	105	**40%**	no ticket evidence — the BA's outside-knowledge pick (label noise)
context_present_but_dropped	103	40%	evidence shown, dropped anyway
weak_broad_context	51	20%	borderline

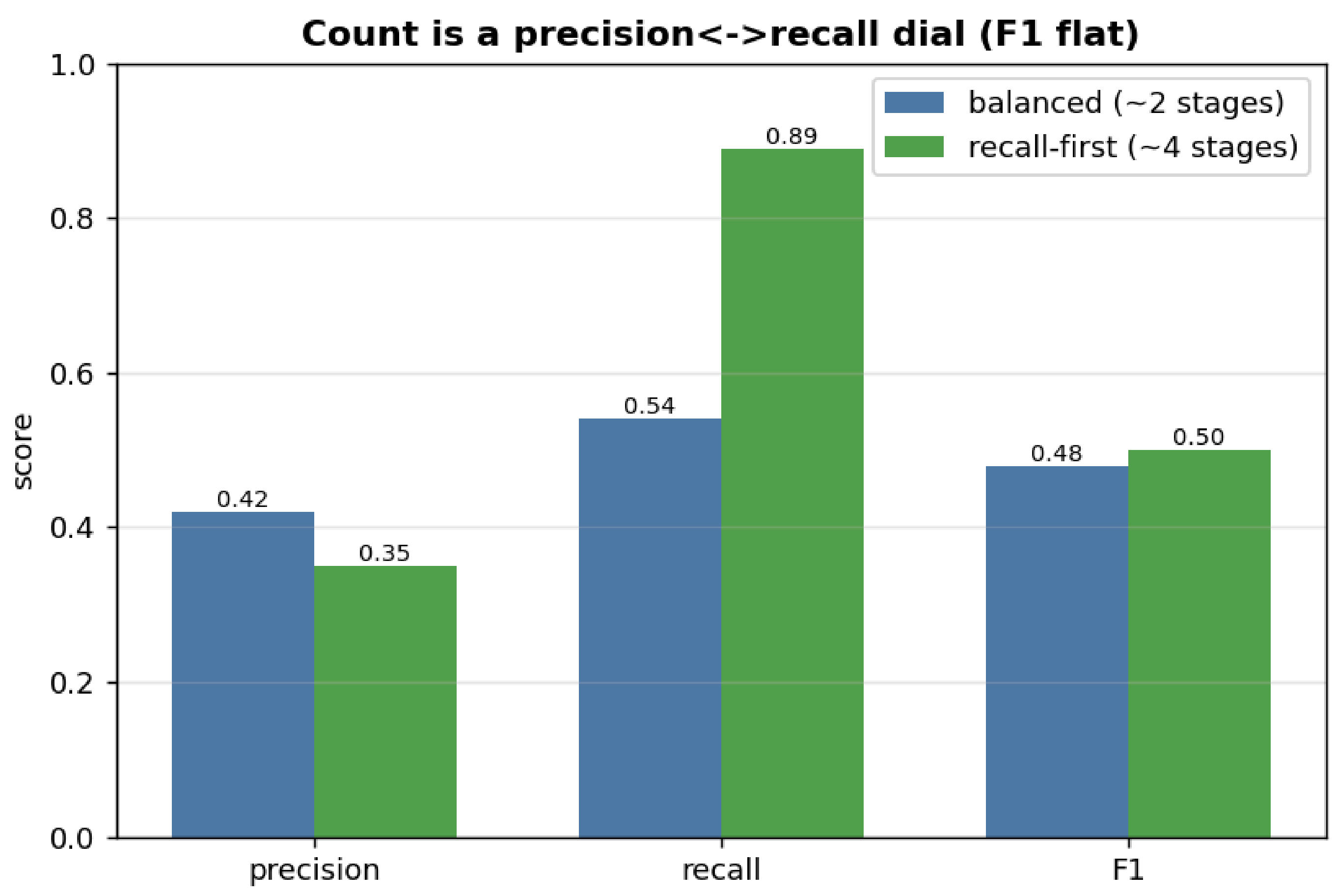


And *why* the picks won over each dropped stage (n = 264):

reason	count	share	read
picks_more_specific	94	35%	the model picked a **better/narrower** stage — often the model being right
no_evidence_for_dropped	69	26%	nothing in the ticket points to the dropped stage
dropped_too_broad	63	24%	the GT stage is broad/adjacent, only loosely implied
adjacent_stage_confusion	28	11%	a neighbouring lifecycle stage picked instead (the one fixable lever)
dropped_is_valid_should_have_picked	**10**	**4%**	**genuine model misses**

So ~40% of drops are un-derivable from the ticket, most of the rest is the model making defensible narrower picks, and only ~4% are genuine misses. The one fixable lever — adjacent-stage confusion — was already pushed down from 17% → 11% with a stage-boundary prompt (pick by entrance/exit scope, and include both adjacent stages when the work spans a boundary).

Finding — the count is a precision↔recall dial (and the GT may be under-tagged)



The model returns a stage count; how generous that is trades precision for recall. Removing the prompt's count guidance entirely makes it a recall-first selector:

config (one_call)	avg predicted	precision	recall	F1
balanced (lean-but-bounded)	~2.0	**0.42**	0.54	0.48
recall-first (no count cap)	~3.9	0.35	**0.89**	0.50

So the same dial seen in value-stream selection applies: the count is a lever, not a fixed answer. **F1 is roughly flat** across the dial (0.48–0.50) — recall-first trades precision ~1:1 for recall.

Crucially, the precision side is probably understated by the ground truth. GT averages only **1.56 stages per VS**, yet a single change often genuinely runs through more of a value stream's lifecycle. The recall-first picks marked "false positives" against that thin GT are frequently *plausible* stages the architect simply did not tag — the same pattern the value-stream judge showed (many non-GT picks rated relevant). So the recall-first config's true precision is higher than 0.35; the strict score penalises it for stages the BA under-tagged or prioritised differently.

This is a product choice, not a bug:

- **recall-first** (no cap) — surfaces every stage the work plausibly touches (~4), recall ~0.89; the architect trims. Best when missing a stage is worse than reviewing an extra.
- **precision-first** (bounded) — a tighter set (~2), recall ~0.54; less to trim. Best when a lean, high-confidence list is wanted.

Both sit at ~0.48–0.50 strict F1; the dial just moves where on the precision/recall curve you operate.

Verdict — locked

Stage selection: raw input + one_call, ~0.48–0.50 strict F1 with a precision↔recall dial (precision-first ~2 stages / recall 0.54, or recall-first ~4 stages / recall 0.89), 0 mislinks.

- **summary slightly beats raw** for stages (~2 pts), but production feeds **raw** for architectural consistency (one input across VS/stages/description/needs) — the cost is within the structural ceiling.
- **one_call** is the production mode: 1 call not N.
- **Interlinking is solved** (isolation prompt + salvage → 0 cross-VS mislinks).
- **The recall ceiling was a catalogue gap**, not the model — pruning uncatalogued GT lifted recall.
- **The count is a precision↔recall dial** — F1 is flat across it (~0.48–0.50); choose recall-first (surface all, architect trims) or precision-first (tight set) per product need. The low precision is partly real over-fetch and **partly thin/under-tagged GT** (1.56 stages/VS, while a change often touches more) — strict precision penalises plausible stages the BA didn't tag.
- **The remaining true gap is structural**: ~40% of drops have no ticket evidence (BA outside-knowledge picks), most of the rest is defensible selection, and only ~4% are genuine misses.

The model performs well against an answerable, noisy target. **No further changes.**